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PAAIM Team Guideline Document

Policy-Aware Agentic Intelligence Manager for Bounded-Autonomy Manufacturing Operations

Internal Guideline v0.1

Prepared for the Research, Product, and Engineering | May 19, 2026

Purpose of this document

To align us around one fundable, publishable, and buildable direction for a new AI manufacturing startup based on PAAIM.

1. Note

This project begins with a simple but powerful observation: factories are already full of intelligence, but that intelligence is fragmented. Cameras detect hazards, quality systems detect defects, maintenance tools predict failures, production systems track throughput, and ERP/MES platforms hold business priorities. Yet when multiple problems occur at once, people still carry the burden of connecting the dots under pressure.

PAAIM is our opportunity to build the missing decision layer for modern manufacturing. The goal is not to replace plant leaders, safety engineers, operators, or maintenance teams. The goal is to give them a governed, explainable, policy-aware AI partner that can turn disconnected alerts into coordinated, auditable, and business-aware actions.

If we execute this carefully, this can become both a high-impact research contribution and the foundation of a fundable AI manufacturing startup. We should build with urgency, but also with discipline: safety first, human governance always, evidence before action, and measurable value in every prototype.

Guiding belief

Manufacturers do not need another dashboard. They need a trusted decision orchestration layer that helps them decide what should happen next when safety, quality, maintenance, production, and cost pressures collide.

2. Top-Level Summary of the Initial Idea

The initial PAAIM idea proposes a policy-driven agentic AI manager that sits above existing industrial AI systems and coordinates specialized agents across safety, quality, maintenance, production, energy, and compliance. Instead of only detecting events and sending alerts, PAAIM reasons about factory policies, resolves conflicts, supports human oversight, and produces explainable decisions.

Initial Strength	Why It Matters
Layer above existing AI systems	Avoids competing with vision, safety, MES, or CMMS vendors; creates an orchestration category.
Multi-agent structure	Maps naturally to manufacturing domains: safety, quality, maintenance, production, energy, compliance.
Policy-aware reasoning	Enables factory-specific behavior instead of one-size-fits-all automation.
Conflict resolution	Directly addresses real plant tradeoffs such as safety vs. throughput and quality vs. speed.
Human oversight and audibility	Makes the concept credible for regulated, safety-sensitive, and enterprise settings.

3. Refined Direction We Should Go For

Refined thesis

PAAIM should be positioned as a policy-aware, bounded-autonomy manufacturing decision orchestration layer - not as an unrestricted autonomous factory controller.

The refined product and research direction should focus on bounded-autonomy incident orchestration. This means PAAIM coordinates decisions under clear policies, known operating limits, approval gates, and evidence requirements. It can recommend, simulate, prioritize, route, and execute only pre-approved low-risk actions. For safety-critical actions, it should work with existing certified safety systems rather than replacing them.

Do This	Avoid This
Position PAAIM as a governed AI operations layer	Position PAAIM as an uncertified AI controller of machines
Start with incident response and recovery	Attempt full autonomous factory management immediately
Show measurable ROI and safety governance	Rely only on futuristic language
Use simulation and digital twin logic for validation	Make claims without controlled experiments
Build a fundable MVP and publishable framework together	Separate research and product tracks too early

4. Startup Vision and Category Creation

The startup category we should aim to create is not computer vision, predictive maintenance, MES, ERP, or digital twin alone. The category is:

Category name

Policy-Aware Industrial Decision Orchestration for Bounded-Autonomy Manufacturing

4.1 One-Line Startup Pitch

PAAIM helps manufacturers move from alert overload to coordinated, policy-compliant action by connecting safety, quality, maintenance, production, energy, and business systems into one governed AI decision layer.

4.2 Why Executives Should Care

- Reduces decision delay during multi-event disruptions.
- Prevents isolated local decisions from harming factory-wide objectives.
- Creates a defensible audit trail for operational decisions.
- Improves downtime recovery, scrap containment, maintenance planning, and labor coordination.
- Enables gradual adoption of AI autonomy without compromising safety or governance.

5. First Product Wedge: Multi-Event Incident Orchestration

The first prototype should focus on a realistic, high-pressure manufacturing scenario where multiple events happen simultaneously and existing systems provide fragmented or contradictory recommendations.

Event Type	Example Signal	PAAIM Response
Safety	Worker enters restricted robot or machine zone	Prioritize safety, check policy, trigger approved workflow, notify supervisor.
Quality	Defect rate rises above baseline	Contain affected batch, inspect root cause, recommend line action.
Maintenance	Vibration indicates bearing degradation	Estimate failure risk and identify service window.
Production	Order target at risk	Estimate quota impact and propose recovery plan.
Energy	High-cost energy window begins	Recommend load shifting when safe and feasible.

5.1 Prototype 0.1 Demo Story

1. Synthetic factory event stream generates simultaneous safety, quality, maintenance, production, and energy signals.
2. Specialist agents analyze each event and produce candidate actions with confidence and evidence.
3. Policy engine applies the plant-specific Industrial Constitution.
4. Decision Twin simulates alternatives and estimates impact on downtime, scrap, OEE, order risk, and safety exposure.
5. Red-Team Agent challenges risky recommendations.
6. Human approval gate routes decisions according to risk level.
7. Dashboard displays recommended action, explanation, approval path, and audit record.

6. Core Product Concepts We Must Preserve

Concept	Definition
Industrial Constitution	A machine-readable set of factory policies defining what AI can do, what requires approval, what is prohibited, and how priorities are resolved.
Bounded Autonomy	AI autonomy is limited by policy, risk, evidence, role permissions, and operational context.
Decision Twin	A simulation and impact-estimation layer that evaluates consequences before recommending actions.
Red-Team Agent	A challenge layer that checks assumptions, sensor reliability, policy violations, and safer alternatives.
Human Approval Gate	A workflow that routes high-impact decisions to the appropriate supervisor, engineer, or manager.
Evidence Pack	A structured record of signals, context, policies, agent reasoning, proposed action, approval, and outcome.

7. System Architecture Guideline

The architecture should be modular so that the first prototype works with simulated data, while future pilots can integrate with real MES, CMMS, QMS, ERP, historian, PLC/SCADA, OPC UA, MQTT, and vision systems.

Architecture Layer	Guideline
Layer 1: Event/Input Layer	Synthetic simulator first; later connect to MES, ERP, QMS, CMMS, historian, sensors, cameras, and energy systems.
Layer 2: Manufacturing Knowledge Model	Assets, lines, cells, products, routings, policies, skills, WIP, materials, shifts, maintenance windows.
Layer 3: Specialist Agents	Safety, quality, maintenance, production, energy, compliance, workforce.
Layer 4: Policy Engine	Priority rules, constraints, approval gates, prohibited actions, evidence requirements.
Layer 5: Decision Twin	Scenario simulation, counterfactual impact, tradeoff analysis, recovery planning.
Layer 6: Governance and Action Layer	Human approval, work order creation, schedule recommendations, notifications, audit logs.
Layer 7: Executive Dashboard	Incident summary, recommended action, ROI impact, risk level, explanation, and traceability.

8. Research and Publication Strategy

Research goal

Publish at least one strong journal article from the initial prototype by making the system experimentally testable, not merely demonstrative.

8.1 First Paper Contribution

The first paper should present PAAIM as a policy-aware multi-agent decision orchestration framework for bounded-autonomy manufacturing operations, validated through simulated multi-event incident scenarios.

8.2 Experimental Baselines

Baseline	Purpose
Alert-only workflow	Represents current fragmented alert systems.
Hardcoded rule workflow	Represents fixed automation without contextual reasoning.
Single-agent LLM	Tests whether one generalist AI is enough.

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Multi-agent without policy engine	Tests the value of explicit policy governance.
PAAIM without Decision Twin	Tests the value of simulation-based impact analysis.
Full PAAIM	Policy engine + specialist agents + Decision Twin + red-team + human gate.

8.3 Metrics to Track

- Correct prioritization rate
- Policy violation rate
- Decision latency
- Downtime avoided or reduced in simulation
- Scrap or defect containment improvement
- Human approval workload
- Quality of explanations
- Robustness under missing or conflicting sensor data
- Recovery plan quality
- Audit completeness

9. Product Development Roadmap

Phase	Timeline	Output
Phase 0: Alignment	1-2 weeks	Finalize thesis, use cases, system boundaries, and prototype scenario.
Phase 1: Prototype 0.1	4-6 weeks	Build event simulator, agents, policy engine, basic Decision Twin, dashboard, audit log.
Phase 2: Research Validation	4-6 weeks	Run simulation experiments, compare baselines, produce results tables and plots.
Phase 3: Executive Demo	2-3 weeks	Create company-customizable demo story, ROI model, funding pitch, and pilot package.
Phase 4: Pilot Readiness	8-12 weeks	Add real connector strategy, security model, role permissions, deployment plan, and partner proposal.

10. Funding and Startup Readiness Guideline

10.1 What Investors and Industry Sponsors Should Hear

- Manufacturers already have AI, sensors, and dashboards, but they do not have a trusted decision orchestration layer.
- PAAIM does not replace existing systems; it increases the value of those systems by coordinating their outputs into action.
- The first product wedge targets costly multi-event disruptions where safety, quality, maintenance, and production priorities conflict.
- The system is built around bounded autonomy, auditability, and human approval - making it enterprise adoptable.
- The prototype produces both product evidence and publishable research evidence.

10.2 Investor-Ready Proof Points to Build

- A working command-center demo using realistic simulated incidents.
- A before/after incident replay showing how PAAIM changes decision quality and recovery speed.
- A measurable ROI model: downtime, scrap, maintenance, safety exposure, energy, and labor coordination.
- A clear integration story with existing manufacturing systems.
- A journal-paper pipeline demonstrating technical credibility.

- A pilot customization checklist for automotive, electronics, medical device, food, and industrial equipment manufacturers.

11. Non-Negotiable Design Principles

Principle	Meaning
Safety First	PAAIM must not bypass certified safety systems or industrial controls.
Human-Governed Autonomy	Every autonomous action must be bounded by explicit policy and role permissions.
Evidence Before Action	Every recommendation must show the signals, context, and assumptions used.
Policy as Product	The Industrial Constitution is a core asset, not an implementation detail.
Simulation Before Action	High-impact decisions should be tested through the Decision Twin before execution.
Audit by Default	Every event, recommendation, decision, approval, action, and outcome must be recorded.
Integration Friendly	PAAIM should sit above existing manufacturing systems, not require replacement.
Measurable Value	Every demo must quantify operational impact.

12. Immediate Sprint Backlog

Priority	Task
Sprint Item 1	Finalize the multi-event incident scenario and define all event types.
Sprint Item 2	Draft Industrial Constitution v0.1 with allowed, approval-required, and prohibited actions.
Sprint Item 3	Design synthetic factory data schema: machines, line, products, events, shifts, policies, KPIs.
Sprint Item 4	Build event simulator that can generate normal, warning, failure, and simultaneous-conflict scenarios.
Sprint Item 5	Implement specialist agents for safety, quality, maintenance, production, and energy.
Sprint Item 6	Implement initial policy engine and decision scoring.
Sprint Item 7	Implement basic Decision Twin impact estimation.
Sprint Item 8	Build dashboard wireframe and approval-gate workflow.
Sprint Item 9	Create audit log schema and evidence-pack format.
Sprint Item 10	Define experimental baselines and metrics for journal paper.

13. Suggested Internal Milestones

Milestone	Name	Completion Signal
Milestone 1	Concept locked	Shared vision, refined thesis, first scenario.
Milestone 2	Paper skeleton ready	Abstract, introduction outline, research gaps, contribution claims.
Milestone 3	Prototype shell ready	Dashboard, event stream, placeholder agents, audit display.
Milestone 4	Policy engine working	Industrial Constitution applied to sample incidents.
Milestone 5	Decision Twin working	Alternative actions scored by downtime, scrap, risk, and OEE impact.
Milestone 6	Full demo working	Multi-event incident orchestrated end-to-end with approval and audit trail.
Milestone 7	Experiment complete	Baseline comparison results generated.
Milestone 8	Funding package ready	Demo video, pitch deck, one-page concept note, pilot customization template.

14. Recommended Manuscript Outline

1. Introduction: manufacturing decision fragmentation and need for bounded autonomy.
2. Related Work: smart manufacturing, digital twins, multi-agent systems, industrial AI governance, human-in-the-loop AI.
3. Problem Formulation: multi-event manufacturing incident orchestration under policy constraints.
4. PAAIM Architecture: agents, policy engine, Decision Twin, red-team, approval gate, audit layer.
5. Prototype Implementation: simulator, system components, UI, data schema, action model.
6. Experimental Design: baselines, scenarios, metrics, evaluation process.
7. Results: decision quality, policy compliance, latency, recovery impact, robustness.

8. Discussion: industry feasibility, safety boundaries, deployment constraints, limitations.
9. Conclusion: contribution, product implications, and path to real-world pilots.

15. Final Alignment Statement

Team commitment

We are building a new class of manufacturing AI system: not a model, not a chatbot, and not another dashboard, but a governed agentic decision layer that helps factories respond intelligently when competing operational priorities collide.

We should move forward with one shared objective: build a credible Prototype 0.1 that is impressive enough for executives, rigorous enough for a top journal article, and practical enough to become the first product of a new AI manufacturing startup.

Appendix A: Company Customization Inputs for Future Demos

- Top three operational pain points.
- Critical assets, bottleneck machines, and production lines.
- Current KPIs: OEE, downtime, scrap, throughput, labor utilization, energy cost.
- Existing systems: MES, ERP, CMMS, QMS, historian, SCADA/PLC, vision AI.
- Safety policies and escalation rules.
- Maintenance windows and shift patterns.
- One historical disruption or incident to replay.
- Decision rights: who can approve what action and under which conditions.

Appendix B: Confidentiality Reminder

This document is intended for internal strategic planning, research development, prototype execution, and funding preparation. Do not distribute externally without permission from the project lead. When sharing externally, remove implementation details, internal roadmap assumptions, and unpublished research claims unless approved.